Lecture 9: Compatible Projections

MATH203 Applied Linear Algebra

Review: What Cross-Filling Discovered

Recall from Lecture 8

We cross-filled a projection $P = P^2$:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_2}$$

We discovered two **remarkable** properties:

1. Each R_i is a projection: $R_1^2 = R_1$, $R_2^2 = R_2$
2. Mutual annihilation: $R_1 R_2 = 0$, $R_2 R_1 = 0$

The Two Properties Together

Property 1 alone is not special — many matrices satisfy $R^2 = R$.

Property 2 alone is not special — many pairs satisfy $AB = 0$.

But **together** they form a remarkable structure:

- Each R_i projects onto its own "floor"
- $R_i R_j = 0$ means: the floor of R_j lies entirely in the **sunlight direction** of R_i

Today: We name this structure, prove a stunning theorem about it, and discover powerful decomposition methods.

Compatible Families

Definition

Definition (Compatible Family)

A collection of projections $\{P_1, P_2, \ldots, P_k\}$ (each satisfying $P_i^2 = P_i$) is called a **compatible family** if:

$$P_i P_j = 0 \quad \text{for all } i \neq j$$

Example: Coordinate Projections in $\mathbb{R}^3$

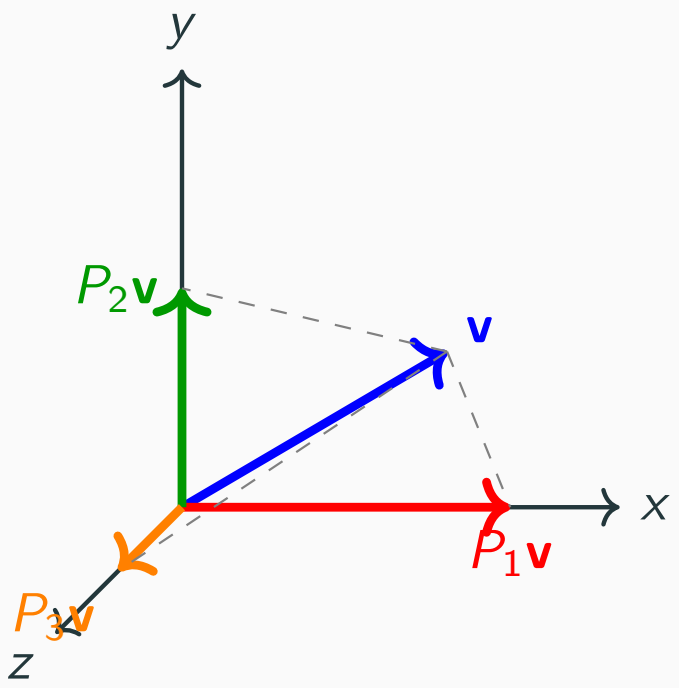
$$P_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Check: each is a projection ($P_1{}^2 = P_1$ ✓, $P_2{}^2 = P_2$ ✓, $P_3{}^2 = P_3$ ✓)

Check: $P_1 P_2 = 0, \quad P_1 P_3 = 0, \quad P_2 P_3 = 0$ ✓

Bonus: $P_1 + P_2 + P_3 = I_3$

Geometric Picture: Coordinate Projections



$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + P_3\mathbf{v}$$

Each projection captures one coordinate. The three "floors" (axes) are completely independent.

Example: From Lecture 8's Cross-Filling

Recall:

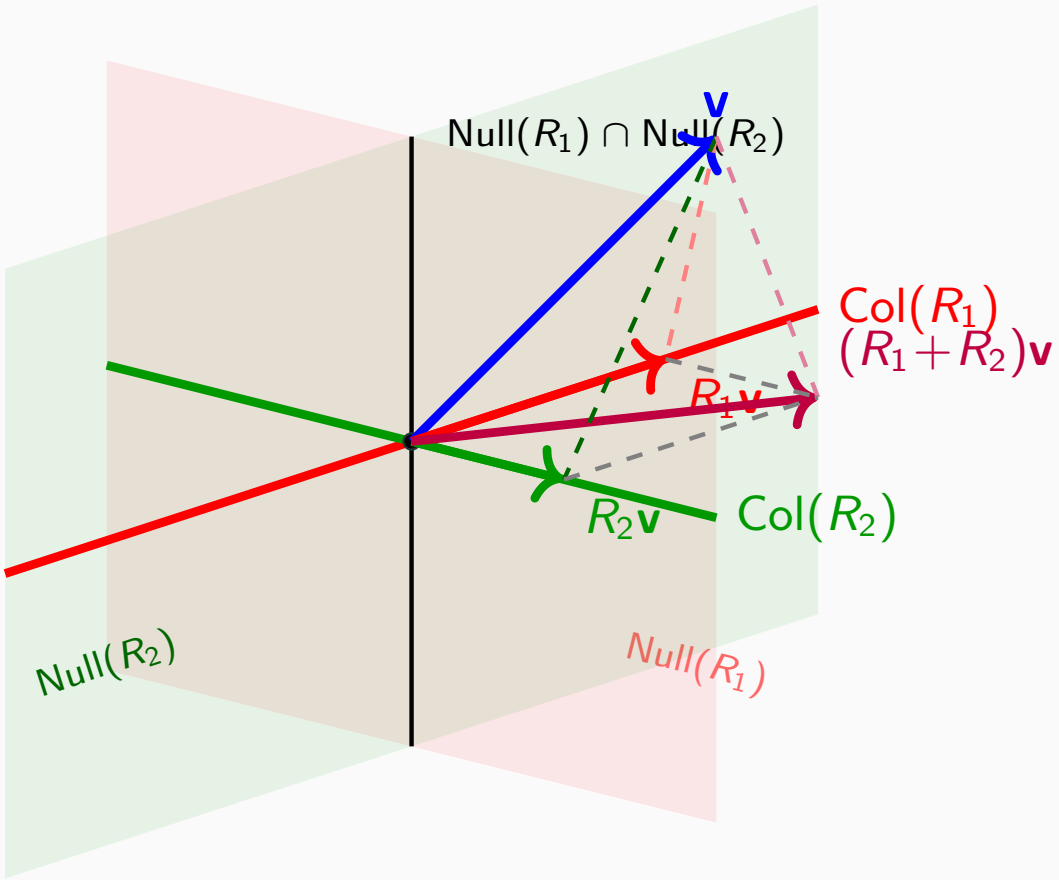
$$R_1 = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

We already verified: $R_1^2 = R_1$, $R_2^2 = R_2$, $R_1 R_2 = 0$.

So $\{R_1, R_2\}$ is a **compatible family**.

This is not a coincidence — we will prove that cross-filling **always** produces compatible families.

Geometric Picture of Compatible Families



Containment (the geometric meaning of $R_1 R_2 = 0$):

- $\mathrm{Col}(R_2) \subset \mathrm{Null}(R_1)$: green line lies inside red plane
- $\mathrm{Col}(R_1) \subset \mathrm{Null}(R_2)$: red line lies inside green plane

Property 1: Subsums Are Projections

Proposition

Let $\{P_1, \ldots, P_k\}$ be a compatible family. Then any subsum

$$Q = P_1 + P_2 + \cdots + P_j$$

is also a projection ($Q^2 = Q$).

Proof: Expand Q^2:

$$Q^2 = (P_1 + P_2 + \cdots + P_j)(P_1 + P_2 + \cdots + P_j)$$

$$= P_1^2 + \underbrace{P_1 P_2}_{=0} + \cdots + P_2^2 + \underbrace{P_2 P_1}_{=0} + \cdots + P_j^2$$

All cross terms vanish ($P_i P_l = 0$), only diagonal terms survive:

$$Q^2 = P_1^2 + P_2^2 + \cdots + P_j^2 = P_1 + P_2 + \cdots + P_j = Q \quad \checkmark$$

Verify with Coordinate Projections

$$P_1 + P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ is a projection } \checkmark \quad \text{(projects onto } xy\text{-plane)}$$

$$P_2 + P_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is a projection } \checkmark \quad \text{(projects onto } yz\text{-plane)}$$

$$P_1 + P_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is a projection } \checkmark \quad \text{(projects onto } xz\text{-plane)}$$

$P_1 + P_2 + P_3 = I_3$ is a projection $\checkmark$

Property 2: Independent Decompositions

Proposition

Let $\{P_1, \ldots, P_k\}$ be a compatible family. For any vector $\mathbf{v}$, the nonzero vectors among $\{P_1\mathbf{v}, \ldots, P_k\mathbf{v}\}$ are linearly independent.

Proof: Suppose $c_1 P_1 \mathbf{v} + \cdots + c_k P_k \mathbf{v} = \mathbf{0}$.

Left-multiply by P_j:

$$c_1 \underbrace{P_j P_1}_{=0} \mathbf{v} + \cdots + c_j \underbrace{P_j^2}_{=P_j} \mathbf{v} + \cdots + c_k \underbrace{P_j P_k}_{=0} \mathbf{v} = \mathbf{0}$$

$$c_j P_j \mathbf{v} = \mathbf{0}$$

If $P_j \mathbf{v} \neq \mathbf{0}$, then $c_j = 0$. $\square$

Unique Decomposition

If $\{P_1, \ldots, P_k\}$ is a compatible family with $P_1 + \cdots + P_k = P$, then every vector $P\mathbf{v}$ decomposes as:

$$P\mathbf{v} = \underbrace{P_1\mathbf{v}}_{\in \mathrm{Col}(P_1)} + \underbrace{P_2\mathbf{v}}_{\in \mathrm{Col}(P_2)} + \cdots + \underbrace{P_k\mathbf{v}}_{\in \mathrm{Col}(P_k)}$$

By Property 2 (linear independence), this decomposition is **unique**.

(In other textbooks, this is called a **direct sum decomposition**, written $\mathrm{Col}(P) = \mathrm{Col}(P_1) \oplus \cdots \oplus \mathrm{Col}(P_k)$. We will not use this notation.)

Key: $P_i P_j = 0$ ensures the "floors" are independent — the floor of P_i lies in the sunlight direction of P_j.

The Projection Decomposition Theorem

The Big Question

Compatible families seem like a very **special** structure. We required:

1. Each $P_i^2 = P_i$ (projection)
2. $P_i P_j = 0$ for $i \neq j$ (mutual annihilation)

Suppose we write a projection as a sum:

$$P = R_1 + R_2 + \cdots + R_k$$

We don't assume the R_i are projections. We don't assume $R_i R_j = 0$.

Question: Under what conditions can we **guarantee** that $\{R_1, \ldots, R_k\}$ is a compatible family?

The Stunning Answer

Theorem (Projection Decomposition — The Climax)

Let P be a projection ($P^2 = P$). Suppose:

$$P = R_1 + R_2 + \cdots + R_k$$

where each R_i is a matrix (not necessarily a projection!), and:

$$\text{rank}(R_1) + \text{rank}(R_2) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$$

Then **automatically**:

1. Each R_i is a projection ($R_i^2 = R_i$)
2. $R_i R_j = 0$ for all $i \neq j$ (compatible family)
3. Equality holds: $\text{rank}(R_1) + \cdots + \text{rank}(R_k) = \text{rank}(P)$

Why Is This Remarkable?

We **never assumed** the R_i are projections.

We **never assumed** $R_i R_j = 0$.

The projection property and mutual annihilation **emerge automatically** from the rank condition alone!

To check compatibility, we only need:

$$\text{rank}(R_1) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$$

Key: The rank condition forces a rigid algebraic structure.

Example: Let's Try It

Take a rank-2 projection (from Lecture 8's diagonal cross-filling):

$$P = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \qquad \text{trace} = \frac{2+2+2}{3} = 2 = \text{rank}(P)$$

Cross-fill at diagonal pivot $(1,1)$, then pivot $(2,2)$:

$$P = \underbrace{\frac{1}{6}\begin{pmatrix} \mathbf{4} & -2 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & 1 \end{pmatrix}}_{R_1,\ \text{rank}=1} + \underbrace{\frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & -1 & 1 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

Red = pivot, blue = cross-arm, black = determined by cross

Example: Check the Rank Condition

$$\text{rank}(R_1) + \text{rank}(R_2) = 1 + 1 = 2 = \text{rank}(P) \quad \checkmark$$

The theorem **predicts** (without any computation!):

1. R_1 is a projection (${R_1}^2 = R_1$)
2. R_2 is a projection (${R_2}^2 = R_2$)
3. $R_1 R_2 = 0$ (compatible)

Quick sanity check (trace = rank for projections, Lecture 8 §3):

$\text{trace}(R_1) = \frac{4+1+1}{6} = 1 = \text{rank}(R_1)$ ✓,
$\text{trace}(R_2) = \frac{0+1+1}{2} = 1 = \text{rank}(R_2)$ ✓

Why Does the Rank Condition Suffice?

The example confirms: the rank condition **works**.

But **why**? Why does a single inequality about ranks force both $R_i^2 = R_i$ and $R_i R_j = 0$?

Time for the proof.

Proof Strategy: Start Simple

I_n is itself a projection: $I_n^2 = I_n$.

Let's first prove the theorem for the **simplest** case $P = I_n$:

$$R_1 + R_2 + \cdots + R_k = I_n, \quad \text{rank}(R_1) + \cdots + \text{rank}(R_k) \leq n$$

Once we handle this case, we will upgrade to a general projection P.

Proof: Step 1 — Factor Each R_i

Write $r_i = \text{rank}(R_i)$. By cross-filling (Lecture 8), each R_i factors as:

$$R_i = \underbrace{U_i}_{n \times r_i} \cdot \underbrace{V_i}_{r_i \times n}$$

Let $s = r_1 + r_2 + \cdots + r_k =$ **total number of columns across all U_i**.

By assumption: $s \leq n$.

Proof: Step 1 (continued) — Assemble Block Matrices

Stack all U_i side by side, all V_i on top of each other:

$$\underbrace{\begin{pmatrix} | & | & & | \\ U_1 & U_2 & \cdots & U_k \\ | & | & & | \end{pmatrix}}_{U,\ \text{size } n\times s} \quad \underbrace{\begin{pmatrix} - & V_1 & - \\ - & V_2 & - \\ & \vdots & \\ - & V_k & - \end{pmatrix}}_{V,\ \text{size } s\times n}$$

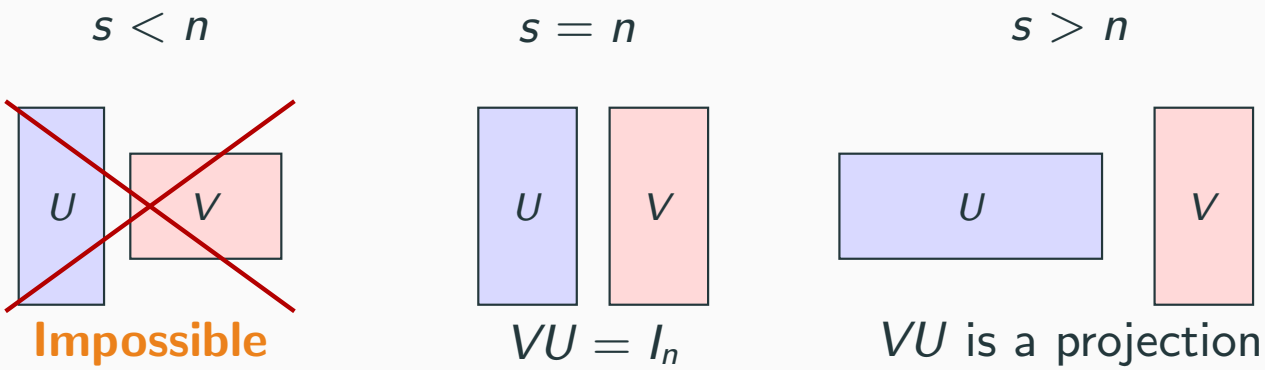
Then:

$$UV = U_1V_1 + U_2V_2 + \cdots + U_kV_k = R_1 + R_2 + \cdots + R_k = I_n$$

Proof: Step 2 — Callback: Three Cases from Lecture 8

Recall (Lecture 8 §5): If $UV = I_n$ (U is $n \times s$, V is $s \times n$), then VU is a projection of rank n. Since rank $\leq$ size: $n \leq s$.

Three cases:



- $s < n$: $\text{rank}(VU) \leq s < n$ — contradicts rank $= n$. **Impossible.**
- $s = n$: rank-n projection on $\mathbb{R}^n \implies VU = I_n$.
- $s > n$: VU is a **genuine** rank-n projection on $\mathbb{R}^s$.

Proof: Step 2 (continued) — Apply to Our Situation

From Step 1: $\underbrace{U}_{n\times s} \cdot \underbrace{V}_{s\times n} = I_n$ and $s \leq n$.

The rank condition gives us $s \leq n$, which **eliminates case 3** ($s > n$).

Case 1 ($s < n$) is always impossible.

Only **case 2** survives: $s = n$.

U and V are both **square** ($n \times n$), and $VU = I_n$.

Proof: Step 4 — What Does $VU = I_n$ Look Like?

Recall the block structure (from Step 1):

$$\underbrace{\begin{pmatrix} - & V_1 & - \\ - & V_2 & - \\ & \vdots & \\ - & V_k & - \end{pmatrix}}_{V} \cdot \underbrace{\begin{pmatrix} | & | & & | \\ U_1 & U_2 & \cdots & U_k \\ | & | & & | \end{pmatrix}}_{U} = I_n$$

Row i of V times column j of U gives block (i, j):

$$VU = \begin{pmatrix} V_1 U_1 & V_1 U_2 & \cdots & V_1 U_k \\ V_2 U_1 & V_2 U_2 & \cdots & V_2 U_k \\ \vdots & \vdots & \ddots & \vdots \\ V_k U_1 & V_k U_2 & \cdots & V_k U_k \end{pmatrix}$$

Proof: Step 4 (continued) — Reading Off the Blocks

$$\begin{pmatrix} V_1 U_1 & V_1 U_2 & \cdots & V_1 U_k \\ V_2 U_1 & V_2 U_2 & \cdots & V_2 U_k \\ \vdots & \vdots & \ddots & \vdots \\ V_k U_1 & V_k U_2 & \cdots & V_k U_k \end{pmatrix} = \begin{pmatrix} I_{r_1} & 0 & \cdots & 0 \\ 0 & I_{r_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I_{r_k} \end{pmatrix}$$

Diagonal blocks: $V_i U_i = I_{r_i}$ (same color $\to$ identity)

Off-diagonal blocks: $V_i U_j = 0$ for $i \neq j$ (different color $\to$ zero)

Proof: Step 5 — Each R_i Is a Projection, and $R_iR_j = 0$

From the **diagonal blocks** ($V_iU_i = I_{r_i}$), compute R_i^2:

$$R_i^2 = (U_iV_i)(U_iV_i) = U_i \underbrace{(V_iU_i)}_{=I_{r_i}} V_i = U_iV_i = R_i \quad \checkmark$$

From the **off-diagonal blocks** ($V_iU_j = 0$ for $i \neq j$), compute R_iR_j:

$$R_iR_j = (U_iV_i)(U_jV_j) = U_i \underbrace{(V_iU_j)}_{=0} V_j = 0 \quad \checkmark$$

This completes the proof for $P = I_n$. $\square$

Proof Summary ($P = I_n$ Case)

The entire proof flows from one chain:

$$R_1 + \cdots + R_k = I_n \xrightarrow{\text{factor}} UV = I_n \xrightarrow{s \leq n} s = n,\ VU = I_n$$

From the **block structure** of $VU = I_n$:

- **Diagonal blocks**: $V_i U_i = I_{r_i} \implies R_i^2 = R_i$ (projection!)
- **Off-diagonal blocks**: $V_i U_j = 0 \implies R_i R_j = 0$ (compatible!)

Key: The rank condition forces $s \leq n$, which forces $VU = I_n$, which forces **everything**.

Proof Strategy: Upgrade to General P

We proved the theorem for $P = I_n$. For a general projection P:

Idea: Add the "missing piece" $R_0 = I_n - P$ to **reduce to the I_n case**.

Since P is a projection:

$$(I_n - P)^2 = I_n - 2P + P^2 = I_n - P = R_0 \quad \checkmark$$

So R_0 is also a projection, with $\text{rank}(R_0) = n - \text{rank}(P)$.

Proof: Step 6 — Reduce P to I_n

Add $R_0 = I_n - P$ to both sides of $P = R_1 + \cdots + R_k$:

$$\underbrace{R_0}_{I_n - P} + \underbrace{R_1 + \cdots + R_k}_{=P} = I_n$$

Check the rank condition for the I_n case:

$$\underbrace{\mathrm{rank}(R_0)}_{=n-\mathrm{rank}(P)} + \underbrace{\mathrm{rank}(R_1) + \cdots + \mathrm{rank}(R_k)}_{\leq \mathrm{rank}(P)\ \text{(given)}} \leq n \quad \checkmark$$

By the I_n case: each R_i (including R_0) is a projection, and $R_i R_j = 0$.

In particular, $R_1, \ldots, R_k$ are projections and mutually annihilating. $\square$

Exercise: Verify the Rank Condition

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1,\ \text{rank}=1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

$\text{rank}(R_1) + \text{rank}(R_2) = 1 + 1 = 2 = \text{rank}(P)$ ✓

By the theorem: R_1, R_2 are automatically projections and $R_1 R_2 = 0$.

We get this **for free** — no need to multiply matrices!

Exercise: When the Rank Condition Fails

Same P, but a **different** decomposition:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_1,\ \text{rank}=2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

$\text{rank}(R_1) + \text{rank}(R_2) = 2 + 1 = 3 > 2 = \text{rank}(P)$

Rank condition **fails**. And indeed, R_1 is **not** a projection ($R_1{}^2 \neq R_1$).

Inner Cross-Filling

What If P Is Given in Factored Form?

The Projection Decomposition Theorem checks a **rank condition** on the pieces.

It never requires the pieces to come from cross-filling P itself!

Suppose P has a **factored form**:

$$P = M_1 M_2 \cdots M_k$$

Idea: Cross-fill a **smaller factor** M_i inside P, and let the theorem guarantee compatibility.

Inner Cross-Filling: Theorem

Theorem (Inner Cross-Filling)

Let P be a projection. Suppose $P = M_1 M_2 \cdots M_k$ where $\text{rank}(P) = \text{rank}(M_i)$ for some factor M_i.

Cross-fill $M_i = Q_1 + Q_2 + \cdots + Q_s$ into rank-1 pieces. Define:

$$S_j = M_1 \cdots M_{i-1}\, Q_j\, M_{i+1} \cdots M_k$$

Then $P = S_1 + \cdots + S_s$ is a compatible family of projections.

Remark: We always have $\text{rank}(P) = \text{rank}(M_1 \cdots M_k) \leq \text{rank}(M_i)$ (rank of product $\leq$ rank of each factor). So "$\text{rank}(P) = \text{rank}(M_i)$" is asking that M_i is the **bottleneck** factor — no other factor squeezes the rank further.

Inner Cross-Filling: Why It Works

Clearly $P = S_1 + S_2 + \cdots + S_s$ (distribute the product over the sum).

For rank: each $\text{rank}(S_j) \leq \text{rank}(Q_j) = 1$, so:

$$\text{rank}(S_1) + \cdots + \text{rank}(S_s) \leq \underbrace{1 + \cdots + 1}_{s} = \text{rank}(M_i) = \text{rank}(P)$$

By the Projection Decomposition Theorem: each S_j is a projection and $S_j S_l = 0$ for $j \neq l$. $\square$

Key: We decompose an **inner factor** M_i rather than P itself. The outer factors are "carried along for the ride."

Example: Setup

Same projection $P = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$, but a **different** factorization:

$$\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}}_{U} \cdot \underbrace{\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}}_{M} \cdot \underbrace{\begin{pmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}}_{V}$$

Here M is **not** the identity! But $\operatorname{rank}(P) = 2 = \operatorname{rank}(M)$ ✓.

Example: Inner Cross-Fill M

Cross-fill $M = \begin{pmatrix} \mathbf{1} & -1 \\ 0 & 1 \end{pmatrix}$ at pivot $(\mathbf{1}, \mathbf{1})$:

$$Q_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$S_1 = UQ_1V = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S_2 = UQ_2V = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

${S_1}^2 = S_1$ ✓, ${S_2}^2 = S_2$ ✓, $S_1S_2 = 0$ ✓

Cross-filled the 2×2 factor M (not P itself) — the theorem guarantees compatibility!

Summary

Key Results

1. **Compatible families** (§1–§2): Projections $\{P_1, \ldots, P_k\}$ with $P_i P_j = 0$. Subsums are projections; vector decompositions are unique.
2. **Projection Decomposition Theorem** (§3): $P = R_1 + \cdots + R_k$ with $\sum \operatorname{rank}(R_i) \leq \operatorname{rank}(P) \implies$ each R_i is a projection and $R_i R_j = 0$ — **automatically**.
3. **Inner cross-filling** (§4): If $P = M_1 \cdots M_k$, decompose a bottleneck factor M_i — the outer factors carry along.

Looking Ahead: Lecture 10

We have powerful tools for **decomposing** projections. But we still haven't answered:

Given a desired floor W and sunlight direction S, how do we construct P?

In Lecture 10:

- A projection is **uniquely determined** by $\mathrm{Col}(P)$ and $\mathrm{Null}(P)$
- Construction formula: $P = B(AB)^{-1}A$
- Orthogonal projection formula: $P = B(B^T B)^{-1}B^T$
- Inner cross-filling of $(B^T B)^{-1} \to$ **orthogonal basis**